

WASP-12b

R-0216

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250102 · created 2026-07-17T00:45:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460677.6807 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.11 +/- 0.015
Transit depth (Rp/Rs)^2	1.21 +/- 0.34 [%]
Orbital Inclination (inc)	86.39 +/- 2.53
Airmass coefficient 1 (a1)	1.0429 +/- 0.0088
Airmass coefficient 2 (a2)	-0.038 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	#1 - [490, 264]
Optimal Aperture	3.96
Optimal Annulus	8.6
Transit Duration (day)	0.0999 +/- 0.0022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.11 $\pm$ 0.015 vs 0.1178 (deviation 0.52 σ)
Duration fit vs published	0.0999 d vs 0.125 d (deviation 11.41 σ)
Mid-time O-C	2.0 min
Depth SNR	7.3
Residual scatter	0.85 %

Light curve

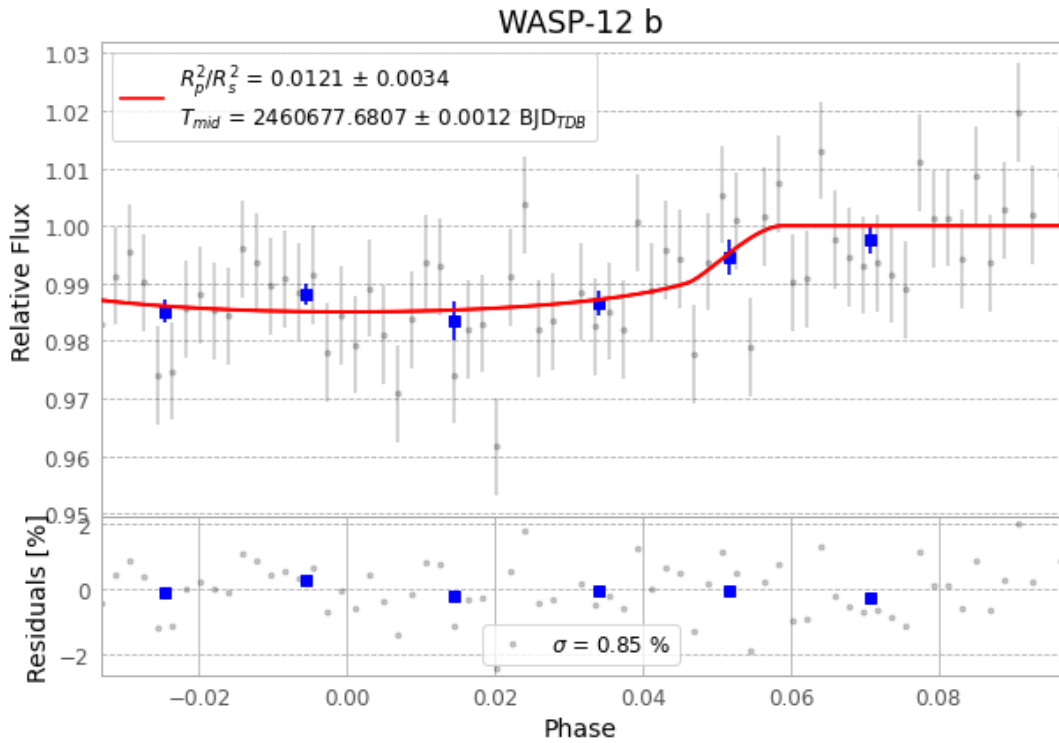


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-01.png (SHA-256 3f2dc8c25fe80b72...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 185], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c3889d14fe1fa504...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

69 FITS frames (45 MB), WASP-12250102031815.FITS → WASP-12250102064215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250102.log (78d96f34f274...), FinalLightCurve_WASP-12 b_2025-01-01.png (3f2dc8c25fe8...), FinalLightCurve_WASP-12 b_2025-01-01.csv (8b8177ab8f6f...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 00:45Z.